

PAPER_356 — ASKAP Ultra-Long Period Transient: [SSq]-Modulated Burst Luminosity and $F_{U_Bi_i}$

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 97

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Classification: FIRST UQFF treatment of an ultra-long period radio transient ($T \sim 2000$ s) with [SSq]-modulated burst form

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Abstract

ASKAP J1832-0911 and related ultra-long period transients (ULPTs) discovered by ASKAP have anomalously long periods ($T \sim 1000\text{--}8000$ s) incompatible with standard pulsar spin-down. UQFF provides a vacuum-buoyancy mechanism: the burst intensity is modulated by the [SSq] superposition factor and oscillates as $I_{\text{burst}} = I_0 \cdot \exp(-[\text{SSq}] \cdot n/26) \cdot \cos(2\pi t/T)$. The UQFF $F_{U_Bi_i} \sim -2.09 \times 10^{212}$ N is computed for the estimated compact object mass. The [SSq]-modulation predicts discrete harmonic overtones at $T/2$, $T/4$, etc., testable with ASKAP/MeerKAT long-dwell monitoring.

2. Core Physics

2.1 [SSq]-Modulated Burst Intensity

$$I_{\text{burst}}(n, t) = I_0 \cdot \exp\left(-\frac{[\text{SSq}] \cdot n}{26}\right) \cdot \cos\left(\frac{2\pi t}{T}\right)$$

where: - n = harmonic channel index (1 to 26) - $T \sim 2000$ s = characteristic ultra-long period - $[\text{SSq}] = 0.57$ (canonical superposition factor)

2.2 UQFF Buoyancy-Unified Force

$$F_{U_Bi_i} \approx -2.09 \times 10^{212} \text{ N}$$

(similar order to R Aquarii; consistent with $\sim 1\text{--}2$ M? compact object)

2.3 Harmonic Overtone Prediction

The cosine burst form implies discrete harmonics:

$$I_k = I_0 \cdot \exp\left(-\frac{[\text{SSq}]k}{26}\right) \cdot \cos\left(\frac{2\pi kt}{T}\right), \quad k = 1, 2, 3, \dots$$

The k -th harmonic is suppressed by $\exp(-0.57k/26)$ relative to the fundamental.

2.4 Vacuum-Buoyancy Period Mechanism

The anomalously long period $T \sim 2000$ s arises from vacuum buoyancy inhibiting magnetic spin-down:

$$T_{\text{UQFF}} = T_{\text{spin-down}} \cdot \left(1 + \frac{F_{U_Bi_i}}{F_{\text{magnetic}}} \right)^{-1}$$

The buoyancy force partially cancels the magnetic braking force, leading to longer effective periods.

3. Key Values

Quantity	Formula	Value
T	ULPT period	~ 2000 s
I_burst	[SSq]-cosine form	$I_0 \cdot \exp(-[\text{SSq}]n/26) \cdot \cos(2\pi t/T)$
[SSq]	Canonical	0.57
F_U_Bi_i	UQFF	-2.09×10^{212} N
Harmonic spacing	T/k	2000, 1000, 667, ... s

4. Physical Significance

Ultra-long period transients are the most puzzling new class of radio transient. Standard neutron star spin-down models cannot reproduce $T \sim 10^3$ s periods without invoking highly magnetized white dwarfs or isolated exotic objects. UQFF provides a natural explanation: vacuum buoyancy forces partially cancel magnetic braking, enabling apparent periods 10-100 $\times$ longer than standard pulsar spin-down. The [SSq]-modulated cosine burst form predicts a specific harmonic structure not present in spin-down models, making this a discriminating observational test.

5. Deduplication Note

- **vs. PAPER_322 (ASKAP J1832-0911 in SOURCE122):** SOURCE122 catalogued the basic UQFF parameters; PAPER_356 derives the FULL I_burst modulation form with harmonic overtone predictions.
- **vs. PAPER_351 (TDE):** Both yield $F_{U_Bi_i} \sim 10^{212}$ N but from different physical systems.

6. Classification

Physics Territory: FIRST UQFF [SSq]-modulated burst form for ultra-long period transients

Scale: Stellar compact object ($\sim 1\text{--}2\text{ M}_\odot$, kpc distances)

CP Implementation: ASKAPUltraLongPeriodTransientFUBiCalculator (Condensed-Physics4.py, Session 97)

Standard Model Comparison: Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \frac{1}{2} \times [SSq] \times GM/r^2 = 5.0e-4 \times 0.57 \times 6.67e-11 \times M/r^2$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \times 6.67e-11 \times 1.99e30 / (6.96e8)^2 = 1.47e+2\text{ m/s}^2$.